

Bella Kim

☎ +1 (519) 722-3297 | ✉ b37kim@uwaterloo.ca | 🌐 bellakdev.vercel.app | 🐙 [GitHub](#) | [in](#) [LinkedIn](#)

EXPERIENCE

Software Engineering Consultant

September 2026 – December 2026

BrainRidge Consulting Inc.

Toronto, ON

- Authored and maintained structured content (**YAML** manifests, **Markdown**-based specifications) for an internal AI agent/skill catalog spanning **15+** functional categories including engineering, QA, legal, and finance
- Enforced **schema validation** against a custom content indexer, resolving manifest and metadata errors across dozens of catalog entries prior to deployment
- Collaborated cross-functionally with engineering and product stakeholders to expand catalog coverage, contributing to a system powering an internal agent store across **3** surfaces (**Claude Code**, **Cowork**, **Copilot Studio**)
- Maintained content versioning and **Git**-based review workflows, ensuring install history accuracy as the catalog scaled to **50+** items

Software Engineering Co-op

January 2026 – April 2026

Graze AI

Remote

- Architected an asynchronous **FastAPI** backend integrating LLM and web-scraping APIs, utilizing **asyncio.Semaphore** to orchestrate **30+** concurrent HTTP requests while strictly managing token rate limits
- Engineered a deterministic entity resolution algorithm implementing fuzzy stem matching, **Levenshtein distances**, and custom heuristic scoring to accurately map unstructured search data to canonical domains
- Developed an automated classification pipeline using **LLMs** to parse natural language criteria into strict **Pydantic** schemas, with fallback deterministic logic to ensure zero-hallucination data mapping
- Implemented a high-throughput caching layer utilizing **aiosqlite** and **SHA-256** fingerprinting, optimizing state management and deduplicating pipeline operations across large-scale CSV datasets

PROJECTS

Autonomous Firefighter Bot | *C++, TraxMaker, AutoCAD, Embedded Systems*

February 2025 – June 2025

- Engineered an autonomous real-time system directly in **C++** to process sensor telemetry, detect anomalies, and execute deterministic pathfinding algorithms within a complex physical maze environment
- Achieved the fastest algorithm execution and physical completion time (**9.8s**) among **50** competing robots
- Designed **3** custom PCBs in **AutoCAD** using TraxMaker to minimize electrical interference, improving hardware-to-software signal reliability and overall system latency by **40%**
- Implemented interrupt-driven motor control and real-time sensor fusion in **C++**, enabling sub-millisecond reaction times to physical environment changes

Hase | *TypeScript, React Native, SQL*

April 2026 – Present

- Architecting a highly concurrent financial tracking client capable of aggregating and logging daily income and expense data streams with **real-time** dynamic visualization
- Engineered a centralized **relational database schema** to securely parse, store, and retrieve digital wallet identifiers and loyalty point metrics across mobile ecosystems

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Computer Engineering

September 2025 – April 2030

- Relevant Coursework: Classical Mechanics, Fundamentals of Programming, Digital Circuits & Systems
- Awards: President's Scholarship of Distinction

TECHNICAL SKILLS

Languages: C++, Python, Java, SQL, TypeScript, JavaScript

Frameworks & Tools: FastAPI, Flask, React, Next.js, Git, SQLite, React Native

Systems & Hardware: Embedded Systems, FreeRTOS, Arduino, PCB Design, AutoCAD, TraxMaker

Markup & Config: YAML, Markdown

Concepts: Asynchronous I/O, ETL Pipelines